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Project history

Field	Value
Version	1.4
Date	2026-10-23
Covers	M0 to M4, the pilot on the client's two companies, PRs #1 to #33

What has been built and done so far, in order: the milestones, every pull request with what it did and why, the problems the pilot found and how each was fixed, the pilot's results, lessons learned and what is still open. Counts and Tally's standard names only: no client names or amounts. The rules themselves are in the rulebook; the database in the schema reference; the decisions in architecture.md (ADR-001 to ADR-038).

1. Milestones

Milestone	What it delivered	PRs
M0 Repository scaffold	Repository layout, docs (requirements, architecture, schema, roadmap), Docker Compose stacks for `test` and `pilot`, `dev.ps1`.	#1
M1 Tally proof of concept	Reading Tally through its XML interface into `raw_records`; the sign convention confirmed on the test company.	#2
M2 Database foundation	The four databases (`ledgerbridge_source`, `_accounting`, `_reporting`, `_system`), database users with least rights, migrations as `migrator`.	#4
M3 Full and incremental sync	Raw records into the canonical tables; AlterID markers; altered, cancelled and deleted vouchers; several companies per run.	#5
M4 Validation	Voucher, trial balance and closing-balance checks after every sync, stored with alerts; then the source's controls (counts, stock, bills, year ends).	#7, then the pilot fixes
Pilot (M9 started early)	The client's two companies loaded from backups in the `pilot` environment; the developer runs it, Claude Code never touches it.	#8 to #33

Design work alongside: AI data protection (#3), the dashboard home page design (#6), budget planning (#16), foreign currency (#32), the plain-language codebase guide (#18) and the generated diagrams (#19).

2. Pull requests in order

PR	Date	What	Why
#1	2026-10-09	M0 scaffold and docs	Starting point.
#2	2026-10-09	M1 Tally proof of concept	Prove Tally can be read and the signs understood.
#3	2026-10-09	AI data-protection requirements and design (AI-011 to AI-018, ADR-015)	Client data must never leave the deployment unprotected.
#4	2026-10-09	M2 schema with readable database names (ADR-016)	One database per layer, least-privilege users.
#5	2026-10-12	M3 full and incremental sync	Canonical model filled from Tally.
#6	2026-10-12	Dashboard home page design (docs)	Agreed design before building M7.
#7	2026-10-12	M4 validation and alerts; pilot preparation	Prove the books are complete and correct.
#8	2026-10-12	`dev.ps1 init-env` path fix; control characters test	First pilot step failed on a path.
#9	2026-10-12	No lock on the connection row during syncs; new connections paused; safe connect and reset (ADR-020)	The pilot's first sync blocked other commands.
#10	2026-10-12	Companies found by GUID; `entity-check`; progress output (ADR-021)	Company names with double and trailing spaces weren't found.
#11	2026-10-12	Chunked, resumable, company-by-company syncs; adaptive splitting (ADR-022)	Large months timed out; a failure lost all progress.
#12	2026-10-12	Unverified chunks never counted complete; part-month probe; refetch from a date (ADR-023)	A part-month listing returned fewer vouchers than the month.
#13	2026-10-13	UDF prefixes; busy source after timeouts; background syncs with `watch` (ADR-024)	Undeclared `UDF:` prefixes broke parsing; closing the terminal stopped syncs.
#14	2026-10-13	`transform retry`, failure breakdowns, company labels, safe data errors, flexible accounting layer (ADR-025)	2,170 ledgers and 238,392 vouchers failed in one batch on a data error.
#15	2026-10-13	`transform rebuild`; only an object's latest version is applied (DAT-008)	Re-convert stored data with new code without Tally.
#16	2026-10-13	Budget planning and analysis in the plan (docs, ADR-026)	Client requirement.

PR	Date	What	Why
#17	2026-10-13	Reference resolution, `transform explain`, voucher type tree, non-posting types (ADR-027)	Records failed on names that differed in spacing or case; custom voucher types.
#18	2026-10-13	Plain-language codebase guide	Onboarding.
#19	2026-10-13	ER, database and data-flow diagrams generated from the models	Keep pictures true to the code.
#20	2026-10-13	`validate analyse`; stock journals, orders and foreign-currency rules	Explain mismatches with counts only.
#21	2026-10-14	Health report, stable error codes, error catalogue (ADR-028)	Readable results for the developer.
#22	2026-10-14	Tally's counting rules reproduced on the test company (ADR-029): openings as at the books' beginning, tracked notes, non-posting vouchers	Ledgers differed by their opening; stock by tracked lines.
#23	2026-10-15	`dev.ps1` arguments, run lock tied to its job, unfinished work first (ADR-030)	`status` split an argument; a lock survived a shutdown.
#24	2026-10-15	Job work and godowns; stock journals read once; the source's own opening difference (ADR-031)	Stock differences on job work; stock journal entries counted twice.
#25	2026-10-15	Stock ledgers with integrated inventory; anonymous ledger profiles	A renamed Stock-in-Hand group's ledger differed.
#26	2026-10-15	Probes for the last ledger differences	Narrow down the remaining differences.
#27	2026-10-15	Compare only on reliable dates in Educational mode (ADR-032)	Tally's balance for the 30th included later vouchers.
#28	2026-10-15	Stock ledgers at the stock value; Tally's own totals; effective dates; narrowed dates (ADR-033)	Two companies reported the stock ledger at the stock value.
#29	2026-10-16	Names as Tally resolves them; deleted masters; line attribution in `validate analyse` (ADR-034)	Three customers differed as if lines sat on the wrong ledger.
#30	2026-10-16	Foreign-currency ledgers at Tally's book value (ADR-035)	The three customers were foreign-currency ledgers, revalued by Tally.
#31	2026-10-16	`validate compare-ledger`; foreign-currency ledgers one by one (ADR-036)	Prove our vouchers equal Tally's, ledger by ledger.
#32	2026-10-16	Foreign currency alongside the base-currency books (docs, ADR-037)	Client requirement: own-currency balances, gains and losses, exposure.

PR	Date	What	Why
#33	2026-10-16	Foreign-currency book value from Tally's vouchers; opening difference as Tally shows it (ADR-038)	Tally's report showed no calculated line to subtract; an opening warning Tally doesn't show.
#34	2026-10-16	Knowledge base: rulebook, generated schema reference, project history, progress log, Word and PDF copies	One place for every rule and table, with proof and history.
#35	2026-10-19	TallyPrime Edit Log checked like for like; seed case and test (no code change)	Confirm the platform works unchanged on Edit Log.
#36	2026-10-19	Security foundations: checks on every pull request, CLAUDE.md security checklist and report sections, threat model and security guide (SEC-009, ADR-039)	Catch bugs, vulnerable dependencies, insecure code and secrets before merging, ahead of M5.
#37 to #44	2026-10-19 to 2026-10-22	dev.ps1 tests on Windows in CI; A3 simplicity review (Tally connector, transform, validation and analyse, workers, scripts) with every result identical before and after; the review's findings fixed	Code the developer can fully understand; bugs found on the way fixed with tests.
#45	2026-10-22	The source's stock values and foreign-currency revaluations stored as source figures; Trial Balance, Profit & Loss and Balance Sheet laid out as Tally's and checked against its own reports line by line (ADR-040)	Reports must equal Tally's; recorded on the test company with integration on and off.
#46	2026-10-22	Educational mode: year-end checks on the 31st or the 2nd, never the 1st (ADR-032 amended)	Tally's computed profit on the 1st disagrees with its ledgers.
#47	2026-10-23	Direct Expenses inside Cost of Sales; the Trial Balance's forex line split into losses and gains; a pilot-safe summary of statement differences	The pilot's statement check failed on both companies (ledger checks all passing).
feat/a5-fx-core	2026-10-24	Foreign currency beside the books (currency, rate, foreign openings, currency masters, rates), the base check after a rebuild, the fx-rules finder (ADR-046)	A5 before M5; find the rule behind ADR-043's stopgap from the pilot's own data.
#54	2026-10-24	Images stamped with their commit; the pilot refuses other code (ADR-045)	Pilot ran an old image after an update.

PR	Date	What	Why
fix/verdicts-alerts-open-items	2026-10-24	Verdicts and exit codes, open items, alert reasons (migration 0019), pilot backups (ADR-044)	Pilot: warnings read as needing action, report exited 1, a stale alert.
#51	2026-10-24	Open differences under their own heading; an alert without a company resolved	Pilot output.
#50	2026-10-24	The Balance Sheet stock fallback retired	Ours matched the pilot at every date.
#49	2026-10-24	Temporary stopgap: Current Assets taken from Tally where ours differs at dates Tally shows its forex line, with a reconciling line; open differences reported (LB-VAL-026), the view `v_statement_open_differences` (ADR-043)	Entity 1 differed at 18 of 78 dates with no rule found.
#48	2026-10-24	The expected company guard (LB-SYNC-011, ADR-041); Trial Balance netting (ADR-042); the test company across a year end; stock ledgers at the source's stock in both settings; lines taken from the source where no rule is known (ADR-043)	Finish statement matching for both pilot companies.
#48	2026-10-24	More tested amounts in the summary (netted split, other stock figures, foreign-currency openings); the report blames the layout when every ledger check passes	The pilot's remaining statement differences.

3. Problems found and how each was fixed

Problem (pilot)	Cause	Fix	PR
Commands blocked during a sync	A transaction held the connection row	Short transactions; commit each batch	#9
Companies not found	Names with double and trailing spaces	Identify companies by GUID	#10
Timeouts and lost progress on large months	One request per month; no checkpoints	Chunks, checkpoints, adaptive splitting, resume	#11
Missing vouchers counted as complete	Part-month listings incomplete	Verify chunks; unverified ranges re-fetched	#12
Parse errors	Undeclared `UDF:` XML prefixes	Prefixes kept and parsed	#13
Thousands of records failed together	One data error failed a whole batch	Record-level failures; values kept in `extra`; retry and rebuild	#14, #15
References not resolved	Spacing, case, aliases, custom voucher types	One resolution rule set; voucher type tree	#17

Problem (pilot)	Cause	Fix	PR
Ledgers off by their opening	Openings fetched for Tally's current period	Fetch as at the books' beginning	#22
Stock counted twice	Invoice lines tracked to notes; stock journals' two lists	Count the note; read stock journals once	#22, #24
Stock differences on job work	A job worker's godown is outside Tally's stock	Godown kinds	#24
Stock ledger differences	Tally reports it at the stock value	Compare with the stock value	#25, #28
Wrong balances on some dates	Educational mode answers only the 1st, 2nd, 31st	Reliable dates	#27
Three customers differing	Foreign-currency ledgers revalued by Tally	Book value from Tally's own vouchers	#30, #31, #33
Opening difference warning	Our figure differed from what Tally displays	Use Tally's own trial balance line	#33
A lock outliving its job; argument splitting	Lock not tied to the job; PowerShell arrays	Lock with heartbeat; argument handling	#23
Statements differing at every date (Profit & Loss 2 lines, Trial Balance 4)	Tally puts Direct Expenses inside Cost of Sales and shows forex losses and gains in separate columns	Same layout, recorded on the seeded test company	#47

4. Pilot results (counts only)

- **Entity 2:** validation **passes**: 415,301 checks, 0 failed.
- **Entity 1:** from 16 failures (Educational-mode dates, PR #27) to 9 (three foreign-currency customers today and at year ends, plus the opening warning). `validate compare-ledger` then matched every one of Tally's vouchers with ours for those customers (47 of 47, 60 of 60, 3 of 3; our extra vouchers were 3 optional Journals each, not counted), with equal totals. PR #33 compares them with Tally's book value and follows Tally's own opening difference; the result of the developer's run after it is to be recorded here.
- **Test company:** passes every check (104 as of 2026-10-02), with the stock-ledger and foreign-currency notes. From 2026-10-24 its books begin on 1 April 2025 (a year end on 31 March 2026): 278 checks, statements equal at all 18 month ends.
- **Statements (after PR #45, 2026-10-23):** every ledger-level check passed on both companies; only the statement check failed: entity 2 (integrated) 6 lines at every date before its first year end, 10 at the year end and the next date, 8 after; entity 1 (not integrated, foreign-currency customers) about 10 at every date. The steady part was the Cost of Sales and forex layout (fix/statement-layout); the year-end part is to be read from the new summary.

TallyPrime Edit Log (2026-10-19)

The test company was opened in TallyPrime Edit Log (migrated, same GUID). Every export the sync and validation use was identical to TallyPrime's in fields, formats and values; an alteration leaves no edit history in the XML export, only a filled `UPDATEDDATETIME` (not fetched). Full sync, validation (105 checks) and the report passed with no code change. A first attempt opened an empty new company of the same name (a different GUID, with job work, cost centres and reversing journals off): nothing was written to it.

5. Lessons learned

- **Reproduce before coding.** Every counting rule was first reproduced on the test company and checked against Tally's own reports (ADR-029); guesses about Tally were wrong more than once (trailing spaces, report periods, revaluation).
- **Count, don't name.** Analysis and reports print counts, kinds and Tally's standard names only, so the pilot could be debugged without sharing client data; detail files stay local (`reports/<env>/private/`).
- **Tally's figures are computed for a period and a date.** Openings, balances, stock and revaluations all depend on the requested period; Educational mode restricts reliable dates.

- **The source's own figure beats our reconstruction.** Opening differences, stock values and foreign-currency book values are taken from what Tally itself shows, with our calculation as a fallback.
- **Make long jobs resumable and visible.** Chunks, checkpoints, background jobs with `watch`, heartbeats and stable error codes made the pilot recoverable.
- **Never guess TDL.** A request with an unknown function froze TallyPrime behind a dialog; only exported field and report names are used.

6. Open items

Item	Status
Sign-off on a licensed TallyPrime	The pilot ran in Educational mode; results must be confirmed on a licensed installation (every date reliable).
Accountant sign-off	An accountant confirms the converted books match the client's (OPEN-001).
Windows Application Control on the developer's machine	Blocks `.venv\Scripts\*.exe`, so `./dev.ps1 test` can't run; tests run with the base interpreter meanwhile.
Foreign-currency data layer	Designed (FX-001 to FX-013, ADR-037), planned for M4b; needs a foreign-currency voucher and a rate entered in TallyPrime first, to copy their exported fields.
Long field paths	The field catalogue keys a field by its path cut to 255 characters, so two paths differing only after that are counted as one (values are unaffected). Left until such paths appear; the fix is a schema change (rulebook Section 12).
Forex revaluation rule (ADR-043)	Entity 1's Current Assets is taken from Tally at 18 dates (open differences, LB-VAL-026) until the rule Tally's Balance Sheet revalues by is found; read-only checks in the client's Tally pending.
Onboarding robustness	Making the first sync of a new company smooth end to end (connect, full sync, validation, report) without manual steps.

Change log

Date	Version	Change
2026-10-23	1.4	PRs #45 and #46; the statement layout fixes and the pilot-safe summary; long field paths.
2026-10-22	1.3	PRs #37 to #44; statements from the source's own figures.
2026-10-19	1.2	PRs #34 to #36; security foundations.
2026-10-19	1.1	TallyPrime Edit Log check.
2026-10-16	1.0	First edition, up to PR #33.